First lessons with MakeCode & the micro:bit workbook

COURSE OVERVIEW

This course presents a sequence of lessons that provide a pathway through six projects, that are perfect to introduce your learners to coding on the micro:bit using Microsoft MakeCode.

We have hand-picked these projects to cover a range of key concepts when exploring coding for the **first time**.

Projects

The following six projects and concepts are explored in this course:

- **Name badge** – introduces making your first algorithm to tell the micro:bit to display a name on the micro:bit's LEDs.
- **Beating heart** – introduces using loops to create beating heart animations.
- **Emotion badge** – introduces inputs and outputs by using two inputs (button A & button B) to trigger two different outputs on the LED display.
- **Step counter** – introduces a variable so that you can record and share the number of steps recorded by the micro:bit.
- **Night light** – uses selection to make different things happen based on different conditions, specifically to decide whether to switch our night light on or not.
- **Rock, paper, scissors** – asks the micro:bit to select a random number and use logic to match it against one of the three possible objects in the game.

A link to our useful glossary.

RESOURCES TO SUPPORT THIS COURSE

This course is supported by the following resources:

- **Introduction videos**
  - Introduction to the micro:bit animation - mbit.io/pd-microbit
  - Introduction to MakeCode video - mbit.io/pd-makecode

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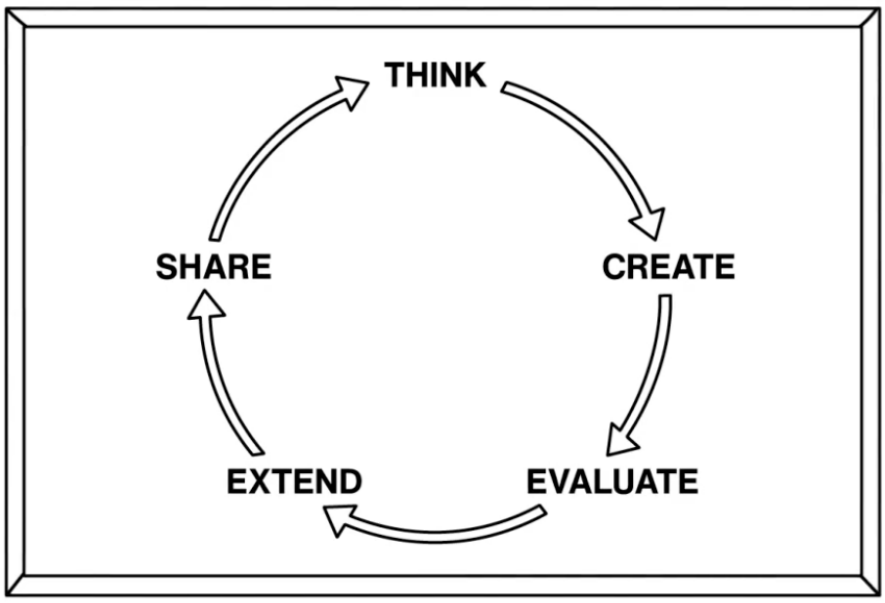
- **MakeCode project tutorials**
  - Name badge - mbit.io/tutorial-name-badge
  - Beating heart - mbit.io/tutorial-beating-heart
  - Emotion badge - mbit.io/tutorial-emotion-badge
  - Step counter - mbit.io/tutorial-step-counter
  - Night light - mbit.io/tutorial-night-light
  - Rock, paper, scissors - mbit.io/tutorial-rps

Downloading code onto your micro:bit

Downloading can vary slightly depending on whether you are using a laptop or a tablet, our transfer code to the micro:bit web page provides the most up-to-date information on this. This will allow you to select and share the method most appropriate to you and your learners.

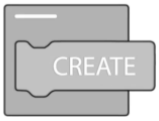
LESSON BREAKDOWN

The next part of this workbook summarises the activity for each project / lesson. It uses the think-create-evaluate-extend-share frame and should act as a quick reminder for what to cover.



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Activity breakdown

LESSON 1 – NAME BADGE	
	**01 THINK** Introduce the micro:bit using the Introduction to the micro:bit video (mbit.io/pd-microbit). Ask your learners to think how they might turn the micro:bit into a name badge. What would it need to do? You are looking for them to spot that they would want to display some text (*their name*) on the LEDs on the front of the device.
	**02 CREATE** Share our short Introduction to MakeCode video (mbit.io/pd-makecode). Decide whether you will lead them through building the program as a whole class or follow the online tutorial (mbit.io/tutorial-name-badge). *We have also shared a code blocks handout for each project in the course download page, which breaks down the code and explains what each block does.*
	**03 EVALUATE** To properly test that the code works, we want to put the code on the device. Follow the download instructions shared in this workbook. Now that the code is on the device, learners can see it working.
	**04 EXTEND** If you want to extend the learning, you can encourage learners to add other strings, perhaps some other facts about themselves. Make sure they test it out and practise transferring the code to their micro:bit.
	**05 SHARE** Bring all the learners together and share both what they have made and what they have learnt during the lesson. Celebrate finishing the first project and let them know they will be creating animations in the next lesson.

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LESSON 2 – BEATING HEART

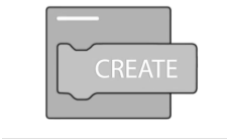


01 THINK

Introduce the concept of animations. Perhaps you can draw on some animation work they have done previously in school such as flip books or stop-frame animations.

Remind learners that animations are a sequence of still images presented with small delays in between that make them appear like they are moving.

Explain they will make some simple animations on their micro:bits using the same principles.



02 CREATE

Decide whether you will lead them through building the program as a whole class or follow the online tutorial (https://mbit.io/tutorial-beating-heart).

We have also shared a code blocks handout for each project in the course download page, which breaks down the code and explains what each block does.



03 EVALUATE

To properly test that the code works, we want to put the code on the device. Follow the download instructions shared in this workbook.

Now that the code is on the device, learners can see it working.



04 EXTEND

If you want to extend the learning, you can encourage learners to animate other things, for example use the diamond or square icons.

Make sure they test it out and practise transferring the code to their micro:bit.



05 SHARE

Bring all the learners together and share both what they have made and what they have learnt during the lesson, especially if they have made their own animations.

Wrap up by letting them know they will be creating an emotion badge in the next lesson.

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LESSON 3 – EMOTION BADGE

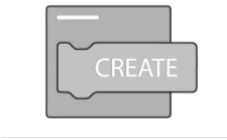


01 THINK

This project is about sharing feelings, so you could start this lesson by exploring different emotions people might have that they would like to share.

In this project, they will code the micro:bit to share happy and sad faces but your learners can extend or change the project to suit them.

You can also use this opportunity to talk about inputs and outputs, as in this project you will be using a button press by the user (*the input*) to trigger an output (*specifically to show a different face or emotion*).



02 CREATE

Decide whether you will lead them through building the program as a whole class or follow the online tutorial (mbit.io/tutorial-emotion-badge).

We have also shared a code blocks handout for each project in the course download page, which breaks down the code and explains what each block does.



03 EVALUATE

To properly test that the code works, we want to put the code on the device. Follow the download instructions shared in this workbook.

Now that the code is on the device, learners can see it working.



04 EXTEND

To extend the learning, you can encourage learners to add other emotions to their badge. There are more emotions in the 'show icon' options including confused, surprised, and angry faces.



05 SHARE

Bring all the learners together and share both what they have made and what they have learnt during the lesson.

Wrap up by letting them know they will be using another input to create their own step counter, or pedometer, in the next lesson.

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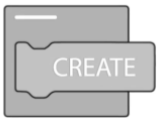
LESSON 4 – STEP COUNTER



01 THINK

Many learners will already be familiar with pedometers and may even own devices such as watches that count their steps. Ask them to think about how they work. You are looking for them to recognise that the device needs to spot movement as a 'step' and then add that step to the running total.

You may also want to run the human step counter unplugged activity with your learners (*available in the course materials download page*). It is a great way for learners to model how a step counter and a variable holding the step total works.



02 CREATE

Decide whether you will lead them through building the program as a whole class or follow the online tutorial (mbit.io/tutorial-step-counter).

We have also shared a code blocks handout for each project in the course download page, which breaks down the code and explains what each block does.



03 EVALUATE

To properly test that the code works, we want to put the code on the device. Follow the download instructions shared in this workbook.

Now that the code is on the device, learners can see it working.



04 EXTEND

To extend the learning, you can encourage learners to investigate the accuracy of their step counter and to consider any improvements they might wish to make to it. Perhaps they want to only show the number of steps when button A is pressed.



05 SHARE

Bring all the learners together and share both what they have made and what they have learnt during the lesson.

Wrap up by letting them know they will be creating a night light in the next lesson.

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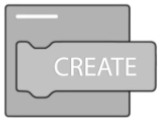
LESSON 5 – NIGHT LIGHT



01 THINK

Ask learners to think about how an automatic night light might work. This is an opportunity to explore selection (*making different things happen based on different conditions).*

In this project, if the light level is below a certain value then switch on the lights, otherwise keep them off.



02 CREATE

Decide whether you will lead them through building the program as a whole class or follow the online tutorial (https://mbit.io/tutorial-night-light).

We have also shared a code blocks handout for each project in the course download page, which breaks down the code and explains what each block does.



03 EVALUATE

To properly test that the code works, we want to put the code on the device. Follow the download instructions shared in this workbook.

Now that the code is on the device, learners can see it working.



04 EXTEND

To extend the learning, you can encourage learners to think about how they might use this project in the real world, perhaps as a way to save electricity or keep people safe in the dark. Encourage them to consider what changes would be needed.



05 SHARE

Bring all the learners together and share both what they have made and what they have learnt during the lesson.

Wrap up by letting them know they will be building on the concept of selection by creating a rock, paper, scissors game in the next lesson.

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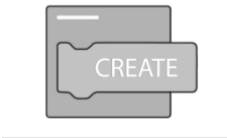
LESSON 6 – ROCK, PAPER, SCISSORS



01 THINK

Play the game in class. This game should be familiar, but it is worth recapping how to play.

Ask learners to spot how the game works. You are looking for them to notice that the choice of object is random.



02 CREATE

Decide whether you will lead them through building the program as a whole class or follow the online tutorial (mbit.io/tutorial-rps).

We have also shared a code blocks handout for each project in the course download page, which breaks down the code and explains what each block does.



03 EVALUATE

To properly test that the code works, we want to put the code on the device. Follow the download instructions shared in this workbook.

Now that the code is on the device, learners can see it working.



04 EXTEND

To extend the learning, you can encourage learners to consider what other games they might code.



05 SHARE

Bring all the learners together and share both what they have made and what they have learnt during the lesson.

Celebrating success and learning is important. Your learners have worked hard over the course of these projects, so this is also an opportunity to reflect on the learning not simply in this lesson but over the six projects.

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AFTER THIS COURSE

Try another course

After you have completed this course and shared the projects with your students, why not try another Professional Development course available through our learning platform.

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Follow us on Twitter @microbit_edu for all our latest news or sign up to our free monthly newsletter which is packed with programming projects and lessons that you can share with your students.

Get in touch if you have ideas for courses

If you have ideas for future course topics, please get in touch with us on info@microbit.org.

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